

■ Working with Multiple Tables in SQL — Revision Notes

■ Subqueries (Query inside another Query)

A **subquery** is a query inside another query — like a mini detective inside your main query! It helps you fetch data based on results from another table.

■ Example 1 — Retrieve employees with jobs listed in the JOBS table:

```
SELECT * FROM EMPLOYEES WHERE JOB_ID IN (SELECT JOB_ID FROM JOBS);
```

■ The inner query finds all valid job IDs, and the outer query fetches employee records that match those IDs.

■ Example 2 — Retrieve jobs for employees earning above 70000:

```
SELECT job_title, min_salary, max_salary, job_id FROM JOBS WHERE job_id IN (SELECT job_id FROM EMPLOYEES WHERE salary > 70000);
```

■ Inner query finds employees with salary > 70000, and outer query lists their job details.

■ Implicit Joins (Old-School Table Connection)

Implicit joins connect two tables using a **WHERE** condition instead of modern JOIN syntax. It's like joining tables manually instead of using the JOIN keyword!

■ Example 3 — Combine employee and job data:

```
SELECT * FROM EMPLOYEES, JOBS WHERE EMPLOYEES.JOB_ID = JOBS.JOB_ID;
```

■ Connects both tables through the **JOB_ID** column to show all related data.

■ Example 4 — Retrieve Department Name for each Employee:

```
SELECT F_NAME, DEP_NAME FROM EMPLOYEES E, DEPARTMENTS D WHERE E.DEP_ID = D.DEPT_ID;
```

■ Connects employees with their departments using department IDs.

■ Quick Recap

Concept	What It Does	Example
Subquery	Query inside another query	SELECT * FROM EMPLOYEES WHERE JOB_ID IN (S
Implicit Join	Connects tables using WHERE	SELECT * FROM EMPLOYEES, JOBS WHERE EMPL

■ Pro Tip: Think of subqueries as 'mini detectives' working inside your main query to uncover the exact data you need!